Assignment 2

Q1. [2+2 Marks] Refer to the table below to answer the following questions:

Movies			
MovieID	**MovieName**	**MovieRating**	**Genre**
M02	Mission Impossible	3.5	Action
M12	It	3.1	Horror
M14	Rush Hour	4.5	Comedy

Actor		
ActorID	**Actor**	**workedOn**
TC1	Tom Cruise	M02
FW3	Finn Wolfhard	M12
JC5	Jackie Chan	M14

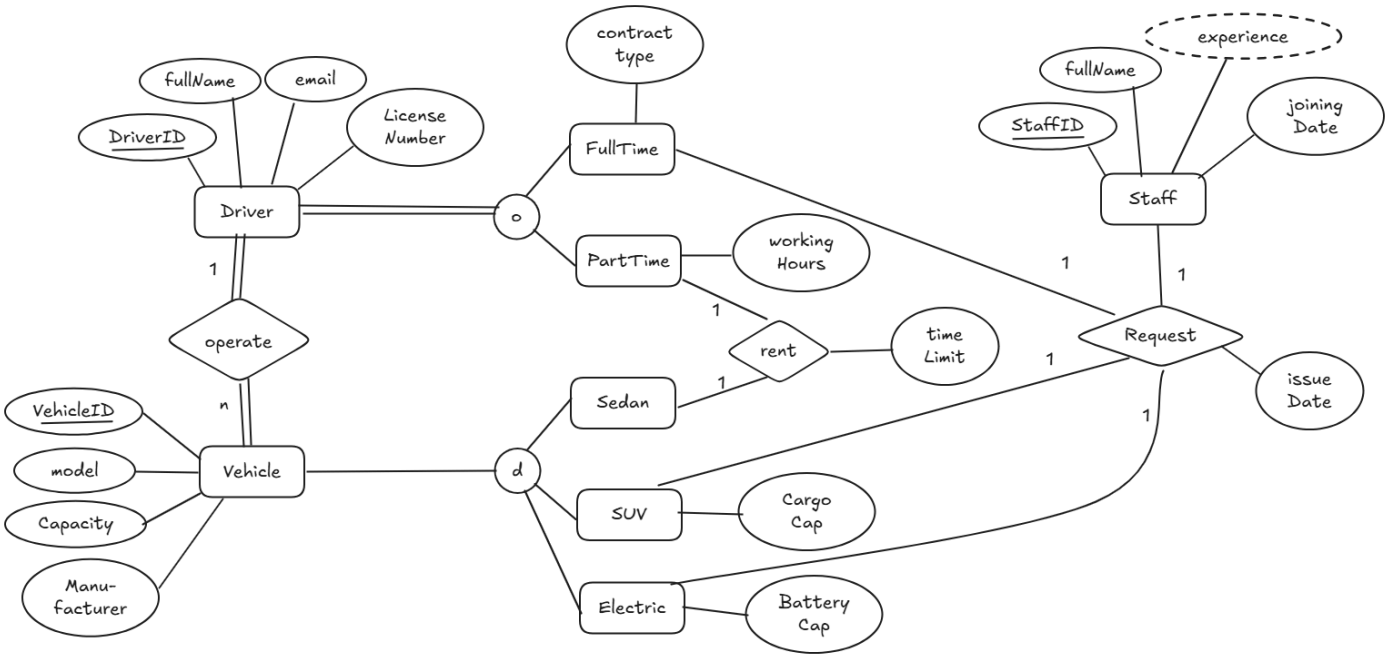
a) Insert <"M12", null, 'X', "Thriller"> into the Movies table.
Identify all constraints which will be violated by this operation and **explain** how you will **solve** it.
b) Insert <null, "Chris Evans", "M13"> into the Actor table.
Identify all constraints which will be violated by this operation and **explain** how you will **solve** it.

a) Ans: Key Constraint violation for M12. Solution: set movieID as PK
Domain Constraint violation for 'X'. Solution: set proper datatype
b) Ans: Entity Integrity Constraint violation for null. Solution: set ActorID as PK
FK Constraint violation for M13, Solution: set proper FK and PK.

Q2. [8 Marks] You are tasked with designing a Driver Management System. The system must meet the following requirements:

- Drivers will have a unique DriverID, full name, email, and license number.
- Each vehicle has a unique VehicleID, model, capacity, and manufacturer.
- Each driver can operate many vehicles, but each vehicle can only be operated by one driver at a time.
- There are only 2 types of drivers: Full-time and Part-time. A driver can be both at the same time. Full-time drivers have a contract_type, while part-time drivers have working_hours.
- Some Vehicles are categorized by their type, such as Sedan, SUV or Electric. A vehicle can belong to only one type. Electric vehicles have BatteryCapacity, SUVs have cargoCapacity.
- A Part-time Driver can rent one Sedan at a time. When a Sedan is rented, the timeLimit is stored.
- A Full-time Driver can request repair from Staff on an SUV or an Electric. The issueDate is recorded when a repair request is made.
- A Staff will have StaffID, FullName, JoiningDate. The experience of a Staff can be calculated from JoiningDate.

Now, draw an **EER** (Extended Entity Relationship) diagram of this system.
Write down all assumptions (if any).



Q3. [8 Marks] Construct a relational Schema by mapping the following EER diagram (Figure Q3). The following conditions must be fulfilled when constructing the schema:

- For mapping the 1:N or 1:1 relationships, you MUST use the foreign key approach.
- For mapping the superclasses/subclasses,
  select the applicable option between 8B or 8C
  and **explain** briefly why the unselected option is not applicable for the given EER.

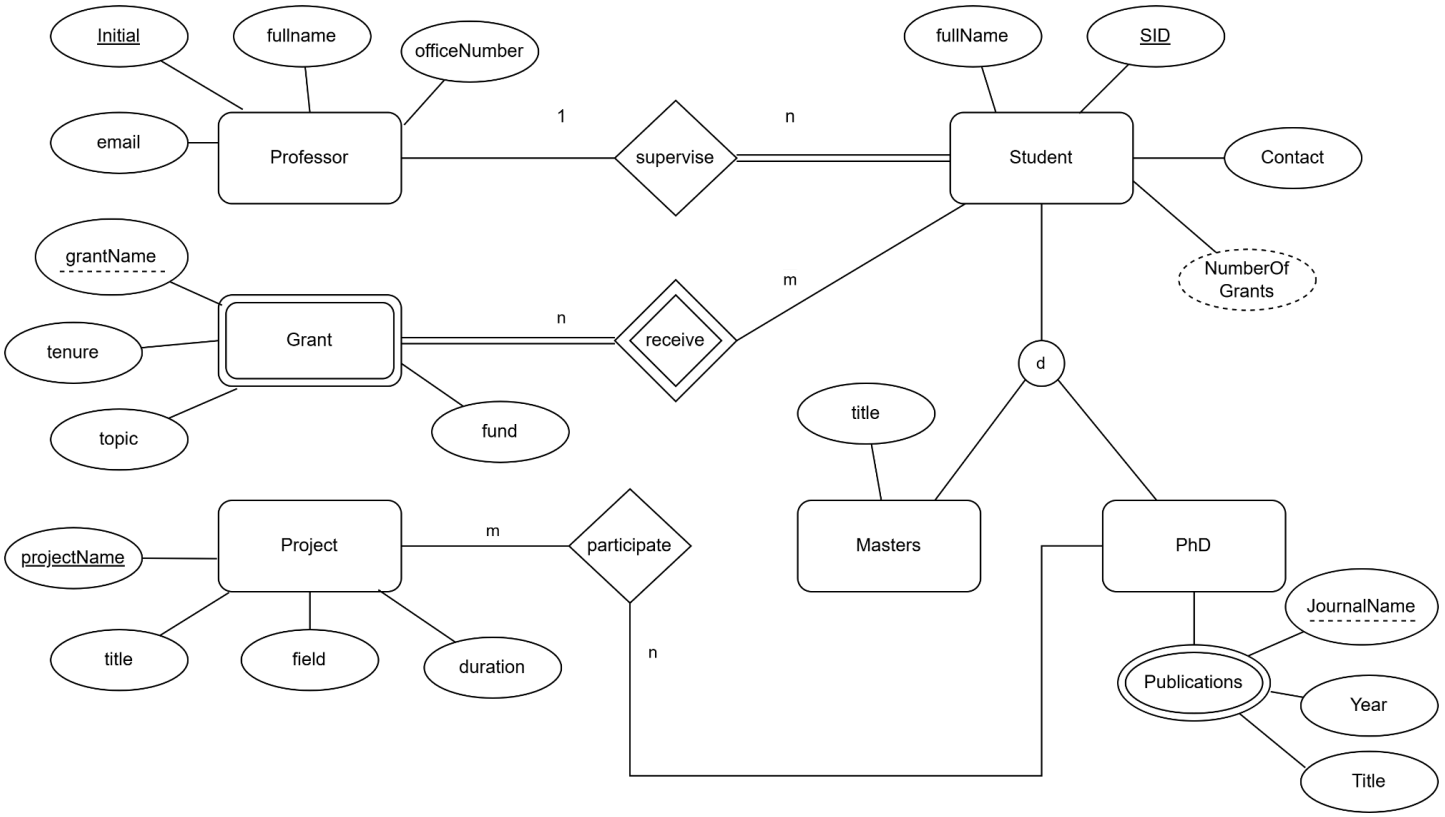
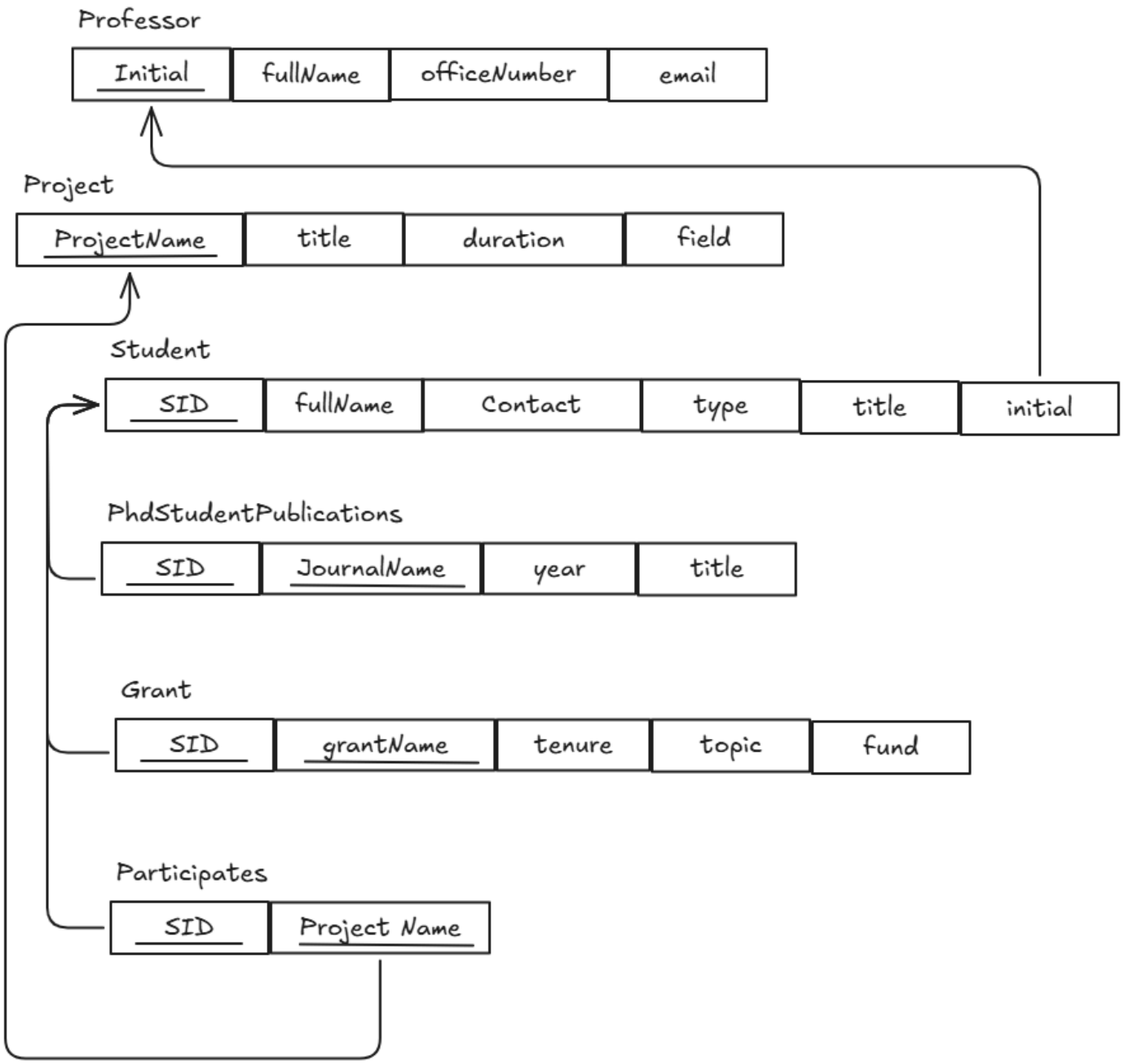


Figure: Q3

Selected 8C

8B can only be used for total.


Professor
Initial
fullName
officeNumber
email
Project
ProjectName
title
duration
field
Student
SID
fullName
Contact
type
title
initial
PhdStudentPublications
SID
JournalName
year
title
Grant
SID
grantName
tenure
topic
fund
Participates
SID
Project Name